

Technical Summary:

Large Language Models are Zero-Shot Reasoners

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1 Research Problem and Motivation

This paper investigates how to elicit multi-step reasoning from large language models (LLMs) in a zero-shot setting for arithmetic, symbolic and logical reasoning tasks. The central research problem is whether a simple, task-agnostic natural language prompt can unlock latent reasoning ability without any task-specific few-shot examples or fine-tuning.

Previous studies on LLMs demonstrated strong performance on single-step or intuition-like tasks such as classification, reading comprehension and translation under few-shot prompting. However, the same models often exhibited poor performance on multi-step reasoning benchmarks such as arithmetic word problems and symbolic reasoning tasks. In those settings, performance curves with respect to model size were almost flat, which indicated that increasing the number of parameters did not automatically translate into better reasoning capability.

Chain-of-thought prompting (CoT) addressed this gap by providing step-by-step reasoning exemplars as few-shot demonstrations. With sufficiently large models such as PaLM, CoT yielded substantial improvements on reasoning benchmarks and restored favourable scaling behaviour. Nevertheless, CoT as originally proposed required carefully hand-crafted examples for each task, and existing work typically emphasised few-shot learning while omitting or under-reporting zero-shot baselines. This situation creates two related gaps:

- It remains unclear whether LLMs already possess strong zero-shot reasoning capabilities that can be elicited by appropriate prompts without any task-specific examples.
- It is not established whether a single fixed natural language template can generalise across diverse reasoning tasks, rather than designing separate prompts for each dataset.

The paper addresses these gaps by proposing a zero-shot chain-of-thought prompting strategy and by systematically comparing it against standard zero-shot prompting and few-shot CoT across multiple benchmarks and model families.

2 Related Work

The study is positioned at the intersection of large language model research, prompt-based learning and reasoning in LLMs.

Large language models such as GPT-3 and PaLM are trained on massive text corpora with autoregressive objectives. After pretraining, these models can be adapted to downstream tasks either through parameter-efficient fine-tuning or through in-context learning. In the in-context setting, a prompt consisting of instruc-

Table 1: Overview of datasets used in the study. Answer formats: N = numeric, M = multiple-choice, Y = yes-or-no, F = free-form text.

Dataset	Domain	Task type	Format	Samples	Notes
SingleEq	Arithmetic	Arithmetic reasoning	N	508	Single-equation
AddSub	Arithmetic	Arithmetic reasoning	N	395	Addition/subtraction
MultiArith	Arithmetic	Multi-step arithmetic	N	600	Multi-step math
GSM8K	Grade school	Multi-step arithmetic	N	1319	Challenging problems
AQUA-RAT	Math MC	Arithmetic with options	M	254	With rationales
SVAMP	Arithmetic	Varied templates	N	1000	GSM-style
CommonsenseQA	Commonsense	Commonsense reasoning	M	1221	World knowledge
StrategyQA	Short Q	Multi-hop reasoning	Y	2290	Implicit strategies
Date Understanding	Temporal	Date reasoning	M	369	Infers dates
Shuffled Objects	Tracking	State reasoning	M	750	Tracks positions
Last Letters	Synthetic	Symbolic reasoning	F	500	Concatenate letters
Coin Flip	Synthetic	Symbolic reasoning	Y	500	Tracks flips

tions and optionally input–output examples is provided, and the model is used to generate continuations that are interpreted as task outputs.

Prompt design has therefore become a central mechanism for controlling LLM behaviour. Manual prompt engineering, automatic prompt search and instruction tuning have been proposed to improve robustness and task generality. A common distinction exists between few-shot prompts that include exemplars and zero-shot prompts that consist only of task descriptions or instructions without explicit examples. The present paper explicitly follows this distinction when designing baselines.

Chain-of-thought prompting is a specific form of few-shot prompting that is tailored to reasoning. Instead of mapping each question to a brief answer, the few-shot examples map questions to multi-sentence rationales that end with the final answer. This approach was shown to improve performance on arithmetic, commonsense and symbolic reasoning, particularly when combined with very large models. However, prior work focussed on few-shot settings and did not explore the possibility that similar reasoning could be elicited in a purely zero-shot fashion.

Related research on reasoning in LLMs also includes fine-tuning methods that train models to output rationales, as well as program-like prompting schemes that decompose complex questions into sequences of simpler sub-questions. These approaches often require additional supervised data, substantial manual engineering or external tools. In contrast, the present study concentrates on prompt design alone and on a single natural language template that is applied across tasks without modification.

3 Dataset Construction

The experimental evaluation covers twelve datasets that together span arithmetic reasoning, commonsense reasoning, symbolic reasoning and other logical reasoning tasks. All datasets are public, with the exception of two synthetic symbolic datasets that are generated by the authors using fixed templates and name lists. Table 1 summarises the main properties of the datasets used in the study.

For Last Letters, instances are generated by sampling four human names from a list and asking the model to output the concatenation of the last letters of these names. For Coin Flip, instances are created by describing a coin that is initially heads up, followed by four people who either flip or do not flip the coin. The task is to answer whether the coin ends in the heads-up state.

All experiments are conducted on evaluation splits only. No additional training data is introduced, and no model parameters are updated, in order to isolate the effect of prompting strategies.

4 Query Protocol and Task Definitions

4.1 Task categories

The tasks are grouped into four main categories.

Arithmetic reasoning. SingleEq, AddSub, MultiArith, GSM8K, AQUA RAT and SVAMP contain arithmetic word problems that require the computation of numeric answers. SingleEq and AddSub mostly involve single-step arithmetic operations such as a single addition or subtraction. MultiArith, GSM8K and SVAMP contain multi-step problems that require a sequence of operations. AQUA RAT is multiple-choice and includes rationales associated with the options.

Commonsense reasoning. CommonsenseQA is a multiple-choice dataset that tests everyday world knowledge, such as typical object locations or functional properties. StrategyQA consists of short yes-or-no questions that require latent multi-hop reasoning strategies, for example combining several facts to determine whether a statement is true.

Symbolic reasoning. Last Letters and Coin Flip focus on symbolic manipulation and state tracking. Last Letters requires concatenation of characters according to a simple rule. Coin Flip requires tracking the state of a coin through a fixed number of possible flips.

Other logical reasoning. Date Understanding and Shuffled Objects originate from the BIG-bench suite. Date Understanding requires temporal reasoning about calendar dates based on textual descriptions. Shuffled Objects requires tracking the positions of objects that undergo a sequence of swaps and answering a question about the final arrangement.

4.2 Prompting regimes

The study compares three main prompting regimes.

Standard zero-shot prompting. In the standard zero-shot setting, the input to the model consists of the question text followed by a simple answer cue. For numeric tasks, a typical template is

Q: [question text]

A: The answer (arabic numerals) is

The model is asked to generate a continuation, and a deterministic answer extraction procedure selects the first valid answer in the required format from the generated text.

Few-shot and few-shot chain-of-thought prompting. Few-shot prompting extends the input with several example question-answer pairs. In standard few-shot prompting, these examples map each question directly to a short answer. In few-shot CoT prompting, the answers in the examples are detailed chains of thought that end with the final answer. The present paper follows existing work in reusing the same few-shot CoT examples, which allows direct comparison.

Zero-shot chain-of-thought prompting. The central contribution is zero-shot CoT, which replaces task-specific examples with a fixed natural language trigger sentence that encourages the model to reason step-by-step. The canonical trigger used in the paper is

Let us think step-by-step.

This trigger is inserted into the prompt in a two-stage pipeline that first elicits a chain of thought and then extracts a final answer.

5 Modeling Approach

5.1 Two-stage zero-shot CoT pipeline

The zero-shot CoT procedure is a two-stage query pipeline.

Stage 1: reasoning extraction. Let x denote the question text, and let t denote the trigger sentence such as “Let us think step-by-step.”. Stage 1 constructs a prompt that contains the question and begins the answer with the trigger. The template has the form

Q: $[x]$. A: $[t]$

The prompt is submitted to the language model, which is decoded using greedy decoding, that is, at each step the most probable next token is selected. The generated continuation, denoted by z , typically consists of multiple sentences that describe intermediate reasoning steps. For arithmetic problems, these sentences restate the known quantities, compute intermediate results and gradually build toward a candidate final answer.

Conceptually, Stage 1 instructs the model to externalise its internal reasoning as natural language text rather than to jump directly to the final answer.

Stage 2: answer extraction. Stage 2 constructs a new prompt that concatenates the original prompt from Stage 1, the generated chain of thought z and an explicit answer cue. For numeric tasks, the cue is of the form

Therefore, the answer (arabic numerals) is

For multiple-choice tasks, the cue is adapted to mention the option format, such as “Therefore, among A through E, the answer is”. The model is queried again with greedy decoding, and the continuation is expected to contain the final answer expressed in the target format.

Intuitively, Stage 2 conditions the model on its own reasoning and forces a concise summary in the form of a single answer. This structure separates the processes of generating a rationale and producing the final prediction.

5.2 Answer cleansing

The raw text produced in Stage 2 can contain multiple numbers, multiple-choice labels or additional words. In order to obtain a single answer for evaluation, the paper uses a deterministic answer cleansing procedure that interprets the output according to the expected answer type.

For numeric answers, the procedure scans the generated text from left to right and extracts the first continuous sequence of digits after removing formatting characters such as commas. This sequence is treated as the model’s predicted number.

For multiple-choice questions, the procedure scans for the first uppercase letter that matches one of the allowed option labels, typically A through E. This letter is taken as the model’s selected option.

For yes-or-no questions, the procedure extracts the first occurrence of either “Yes” or “No”. For free-form answers, the entire generated text is used, optionally after basic truncation. The same cleansing strategy is applied to both standard zero-shot and zero-shot CoT outputs to ensure a consistent evaluation protocol.

5.3 Model families and hyperparameters

The experiments cover a variety of LLM families and sizes:

- Instruction-tuned GPT-3 variants such as text ada 001, text babbage 001, text curie 001, text davinci 001 and text davinci 002.
- Original GPT-3 models of varying parameter counts, including small, medium and large configurations.
- PaLM models with 8B, 62B and 540B parameters.
- Additional models used in scaling analyses, such as GPT-2, GPT-Neo, GPT-J, T0 and OPT.

All models are used in their pretrained form without additional gradient-based updates. The core experiments employ greedy decoding for both stages of zero-shot CoT and for standard zero-shot prompting. Few-shot CoT experiments reuse fixed example sets introduced in earlier work, and the order of examples is held constant across runs.

6 Empirical Results

6.1 Overall zero-shot accuracy improvements

The primary comparison is between standard zero-shot prompting and zero-shot CoT applied to the same models and datasets. Accuracy is measured as the proportion of test questions for which the cleansed prediction exactly matches the ground truth answer.

On arithmetic reasoning tasks, zero-shot CoT yields large gains for problems that require multi-step reasoning. For MultiArith, standard zero-shot prompting on text davinci 002 attains 17.7 percent accuracy, while zero-shot CoT increases this to 78.7 percent. For GSM8K, accuracy increases from 10.4 percent under standard zero-shot prompting to 40.7 percent under zero-shot CoT. On AQUA RAT, accuracy improves from 22.4 percent to 33.5 percent, and on SVAMP it increases from 58.8 percent to 62.1 percent. In contrast, for SingleEq and AddSub, which predominantly contain single-step arithmetic problems, zero-shot CoT offers only marginal changes relative to the already high zero-shot baseline.

On symbolic reasoning tasks, the improvements are even more striking. For Last Letters, standard zero-shot prompting essentially fails, with approximately 0.2 percent accuracy, whereas zero-shot CoT reaches 57.6 percent. This indicates that the model can reliably follow deterministic string manipulation rules when encouraged to reason step-by-step. For Coin Flip, accuracy increases from 12.8 percent to 91.4 percent, which shows that explicit reasoning supports robust tracking of state changes across multiple operations.

On other logical reasoning tasks, zero-shot CoT also improves performance. On Date Understanding, accuracy increases from 49.3 percent to 67.5 percent, and on Shuffled Objects it increases from 31.3 percent to 52.4 percent. These gains illustrate that temporal and object-state reasoning can benefit from explicit chain-of-thought prompts.

Commonsense reasoning results show a mixed pattern. On StrategyQA, which consists of questions requiring latent multi-hop reasoning, accuracy increases from 12.7 percent to 54.8 percent under zero-shot

CoT. In contrast, on CommonsenseQA, accuracy decreases slightly from 68.8 percent to 64.6 percent. Qualitative analysis in the paper notes that, even when the final answer is incorrect on CommonsenseQA, the generated chains of thought are often plausible and reflect reasonable world knowledge.

6.2 Comparison with few-shot prompting and fine-tuned models

The study also compares zero-shot CoT with stronger baselines that use additional information. On MultiArith, standard zero-shot prompting achieves 17.7 percent accuracy, while standard few-shot prompting with two or eight examples yields approximately 33.7 to 33.8 percent accuracy. Zero-shot CoT achieves 78.7 percent under the same model. Few-shot CoT with eight examples further improves performance to 93.0 percent on MultiArith, which demonstrates that explicit examples with rationales still provide additional benefits beyond zero-shot CoT.

On GSM8K, standard zero-shot prompting reaches 10.4 percent accuracy, and standard few-shot prompting reaches 15.6 percent. Zero-shot CoT improves this to 40.7 percent. Few-shot CoT with eight examples achieves 48.7 percent accuracy. The paper also reports a setting where zero-shot CoT and few-shot CoT are combined into a mixed prompt, which yields approximately 51.5 percent accuracy on GSM8K.

These results are compared to a fine-tuned GPT-3 model with 175B parameters trained together with a verifier model, which achieves about 55 percent accuracy on GSM8K. The study further reports that, when applying zero-shot CoT with self-consistency decoding to PaLM 540B, GSM8K accuracy reaches 70.1 percent. Self-consistency decoding samples multiple diverse chains of thought using stochastic decoding and then selects the most frequent final answer among them. This procedure shows that prompt-based methods combined with sampling can outperform previously reported fine-tuned baselines without additional task-specific training.

6.3 Scaling behaviour

The paper analyses how model size affects zero-shot reasoning performance under different prompting strategies. When standard zero-shot prompting is used on arithmetic benchmarks such as MultiArith and GSM8K, performance curves as a function of model size remain almost flat across a wide range of model scales. Larger models do not significantly outperform smaller ones in these settings.

In contrast, when zero-shot CoT is applied, accuracy increases steadily with model size. For both original GPT-3 models and instruction-tuned GPT-3 variants, small models derive little benefit from zero-shot CoT and can even degrade relative to standard zero-shot prompting. However, once the model size exceeds a certain threshold, zero-shot CoT yields substantial gains, and the benefits grow with scale. A similar pattern is observed for PaLM models, where the 8B version shows limited benefit from zero-shot CoT, while the 62B and especially the 540B versions exhibit large improvements.

This behaviour suggests that large LLMs develop internal representations that support multi-step reasoning, but these capabilities are underutilised when the model is queried with naive prompts. The chain-of-thought trigger serves as a mechanism to activate these capacities in a zero-shot regime.

6.4 Prompt template robustness

The authors examine the sensitivity of zero-shot CoT to the exact wording of the trigger sentence by testing multiple variants on MultiArith. Several instructive templates that explicitly encourage reasoning, such as “First,” or “Let us think about this logically.”, produce performance improvements that are comparable but slightly below the default trigger. Templates that decompose the problem into steps or mention logical reasoning generally achieve accuracies above 70 percent, while the phrase “Let us think step-by-step.” yields the highest accuracy of 78.7 percent on the same dataset.

In contrast, misleading templates that discourage reasoning or suggest irrelevant operations, such as “Do not think. Just feel.” or prompts that ask for counting characters in the question, do not improve performance and often revert to standard zero-shot levels. Completely irrelevant sentences that do not pertain to reasoning also fail to provide benefits. These observations indicate that the performance gains arise from semantically meaningful instructions to perform step-by-step reasoning, rather than from arbitrary changes in the prompt text.

6.5 Few-shot CoT robustness and error analysis

The robustness of few-shot CoT is evaluated by mixing examples and target tasks. When chain-of-thought examples from CommonsenseQA are used as demonstrations for AQUA RAT, which shares the multiple-choice format but differs in domain, performance still improves over standard zero-shot prompting. However, this performance remains below that of zero-shot CoT, which does not rely on any examples. When examples and target tasks have different answer formats, such as using CommonsenseQA examples for numeric MultiArith problems, the gains become small. These findings support the view that few-shot in-context learning is heavily influenced by answer-format patterns.

The paper also presents qualitative error analysis. On arithmetic tasks, both zero-shot CoT and few-shot CoT produce coherent reasoning chains when the answer is correct. When zero-shot CoT fails, the chain of thought often contains an initially correct intermediate result that is later overwritten by an unnecessary extra computation. By contrast, errors under few-shot CoT more frequently arise from incorrect execution of specific arithmetic operations. On commonsense tasks such as CommonsenseQA, zero-shot CoT often produces plausible multi-step explanations but hesitates between multiple options or fails to commit to a single choice, which results in incorrect final answers despite reasonable rationales.

7 Summary

The paper demonstrates that large language models can serve as effective zero-shot reasoners when prompted with a simple chain-of-thought instruction. A two-stage prompting pipeline that first elicits natural language reasoning and then extracts a final answer yields substantial accuracy improvements on arithmetic, symbolic and logical reasoning benchmarks without any task-specific training or examples.

Relative to previous work, the main contributions are as follows. First, the study introduces zero-shot CoT as a universal prompting scheme that uses a single natural language trigger to elicit step-by-step reasoning across a wide variety of tasks. Second, it establishes that zero-shot CoT dramatically improves zero-shot accuracy on difficult reasoning benchmarks and restores favourable scaling behaviour as model size increases, especially for large models. Third, it provides a systematic analysis of prompt template robustness, cross-task few-shot CoT transfer and error patterns, clarifying how different prompting strategies interact with latent reasoning capabilities in LLMs.

The approach has several limitations. The benefits of zero-shot CoT are concentrated in large models; smaller models do not gain from the chain-of-thought trigger and can degrade in performance. On some commonsense tasks, accuracy does not improve or may decrease, even though the generated rationales appear qualitatively better. The method relies on proprietary pretrained models with limited transparency regarding training data, and it does not address issues of bias, calibration or faithfulness in the produced reasoning. Furthermore, the choice of trigger sentence remains manually engineered, and the optimal design of such prompts is not theoretically characterised.

Future research directions include automated discovery of effective reasoning prompts, integration of verifier models or consistency checks to filter flawed chains of thought, extension of zero-shot CoT to multimodal and interactive reasoning tasks, and more rigorous evaluation of the alignment between generated

reasoning and human standards of correctness and interpretability.